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Windows PowerShell
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PS C:\Users\chang\Desktop\pj\Part5_Alpha\Alpha2\upload_test> iverilog -o core_tb.vvp core_tb.v core.v corelet.v mac_array.v mac_row
.v mac_tile.v mac.v l0.v ofifo.v sfu.v fifo_depth64.v fifo_depth8.v fifo_mux_16_1.v fifo_mux_8_1.v fifo_mux_2_1.v sram_32b_w2048.v
PS C:\Users\chang\Desktop\pj\Part5_Alpha\Alpha2\upload_test> vvp core_tb.vvp
VCD info: dumpfile core_tb.vcd opened for output.
No.           0 execution completed.
No.           1 execution completed.
No.           2 execution completed.
No.           3 execution completed.
No.           4 execution completed.
No.           5 execution completed.
No.           6 execution completed.
No.           7 execution completed.
No.           8 execution completed.

============ PE Gating Statistics (kij=          8) ============
Sample PE[row1][col1] as representative:
  Execute cycles:           52
  Gated cycles:             30
  Gating ratio:             57%
========================================================

============ PE Gating Functional Verification ============
[PASS] Gating logic triggered (gated_cycles =         30 > 0)
===========================================================
```


```
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===========================================================
```


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=================== Gating Analysis ===================

  [Cycle Statistics]
    Total execute cycles:             52
    Normal compute cycles:            22  (gating_a & gating_b = 1)
    Gated cycles:                     30  (gating_a or gating_b = 0)

  [What happens during gated cycles]
    - Multiplier inputs: held at last valid value
    - Output: bypassed to c_q (skipping multiplier result)
    - Multiplier internal switching: reduced (not zero)

  [Important Notes]
    - Gating ratio         57% means         57% of cycles were gated
    - This does NOT equal power reduction percentage!
    - Actual power savings require synthesis + power analysis
    - Tools needed: Design Compiler + PrimeTime PX / Innovus
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```


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```
############ Verification Start during accumulation #############
 1-th output featuremap Data matched! :D
 2-th output featuremap Data matched! :D
 3-th output featuremap Data matched! :D
 4-th output featuremap Data matched! :D
 5-th output featuremap Data matched! :D
 6-th output featuremap Data matched! :D
 7-th output featuremap Data matched! :D
 8-th output featuremap Data matched! :D
 9-th output featuremap Data matched! :D
10-th output featuremap Data matched! :D
11-th output featuremap Data matched! :D
12-th output featuremap Data matched! :D
13-th output featuremap Data matched! :D
14-th output featuremap Data matched! :D
15-th output featuremap Data matched! :D
16-th output featuremap Data matched! :D
############ No error detected ##############
########### Project Completed !! ############
core_tb.v:473: $finish called at 2331000 (1ps)
```